Single splits, phase of care and MRI radiomics in post-treatment glioma surveillance: re-evaluation of a radiomics risk model and an auditable clinical alternative

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Abstract

Background. Models that predict glioma progression from post-treatment MRI are often judged on one held-out patient split. A radiomics model in an internal report (the PRISM report model) scored AUC 0.80 on one 30-patient split of the MU-Glioma-Post cohort.

Purpose. To test whether that performance survives other splits, censoring-aware labels and a clinical comparator, and whether MRI adds information once the phase of care is known.

Methods. Retrospective study of MU-Glioma-Post (308 post-treatment index scans, 155 patients; grade 4: 249 scans, 126 patients) with locked external evaluation on LUMIERE (132 scans, 61 patients). Outcome: documented progression within 120 days of the scan, death censored, scored by IPCW AUC(120 d). We re-scored the PRISM report model on its own split and over 500 splits of the same design, and compared a clinical + timing + as-of-scan treatment ridge model with radiomics (70 configurations, nested selection) and frozen deep features under 5×5 patient-grouped cross-validation. Phase-of-care, label, re-segmentation, diffusion and audit-layer analyses had pre-specified rules.

Results. On the PRISM report split the PRISM report model scored 0.802 and a model without imaging 0.871; with treatment dates leaking from the future, a boosting model reached

0.901. Over 500 splits the PRISM report pipeline averaged 0.720 ± 0.055, and 7% of splits reached 0.802. Under repeated cross-validation the clinical model reached 0.698 (95% CI 0.615–0.774) in grade 4. Radiomics changed this by −0.032 (−0.084 to +0.017) and deep features by −0.004 (−0.069 to +0.063). In post-radiotherapy scans, imaging added +0.061 (−0.038 to +0.156; permutation p = 0.21). Locked external AUCs were 0.608–0.620. An audit layer met its coverage target (0.841, 0.790–0.885), did not improve discrimination beyond case mix, and exposed a missingness shortcut.

Conclusion. The 0.80 reflected a favourable split. Phase of care, not MRI texture, carried the measurable signal. Report repeated cross-validation, phase strata and a clinical comparator.

Key points

- **Question.** Does MRI radiomics or deep learning add to clinical and treatment-phase information for predicting documented progression within 120 days of a post-treatment glioma MRI, and how far can one patient split mislead?
- **Findings.** On one favourable 29-patient split, models scored 0.80–0.90; the same pipelines averaged about 0.70 under repeated cross-validation, and the radiomics pipeline averaged 0.720 over 500 splits of the same design. Radiomics, frozen deep features and re-segmentation added no measurable discrimination to a clinical + treatment-phase model (AUC 0.698 in grade 4), internally or on an external cohort.
- **Meaning.** Single-split AUCs on cohorts of this size measure the split. Imaging-model AUCs that mix pre-radiotherapy and surveillance scans are inflated by phase of care. A clinical model with an audit layer can state its uncertainty and its weak points, but it does not triage scans better than case mix does.

Clinical relevance statement

At a surveillance MRI after glioma treatment, a model built from age, molecular markers and where the scan sits in the treatment course predicts documented progression within 120 days about as well as any MRI-based model tested here (AUC about 0.70). Conventional MRI radiomics should not be expected to add to it, and a single reported test-split AUC near 0.80 should not be taken at face value.

1 Introduction

Most patients with high-grade glioma progress after surgery and chemoradiotherapy. They are followed with MRI every few months. Each surveillance scan raises the same question: how likely is it that progression will be documented before the next decision point? The question is hard for two reasons. First, enhancement can increase after radiotherapy without tumour growth. This pseudoprogression is most common in the first 12 weeks after chemoradiotherapy and is more frequent with MGMT promoter methylation (Brandes et al., 2008; Taal et al., 2008). The RANO criteria therefore treat early post-radiotherapy changes with caution (Wen et al., 2010), and RANO 2.0 asks for confirmation of progression within the first 12 weeks (Wen et al., 2023). Second, "progression" in routine data is a clinical documentation event. It depends on imaging, on symptoms and on when the next scan is scheduled. A model that predicts it is a prognostic model for a pragmatic endpoint, and its intended user is the neuro-oncology team deciding whether to shorten the interval to the next scan or to act on an equivocal finding. For that use, a well-calibrated probability with an honest statement of uncertainty matters as much as a high AUC.

Radiomics and deep learning have been proposed to answer this question from the images. Most work addresses an adjacent task, the separation of pseudoprogression from true progression at a lesion that is already suspicious. Here pooled AUCs are about 0.77 for machine-learning models (Booth et al., 2022), and diffusion and perfusion measures reach AUC 0.90–0.93 in meta-analysis (Taylor et al., 2022). Forward prediction from every surveillance scan has received less attention. On the public MU-Glioma-Post dataset (Mahmoud et al., 2025), a calibrated radiomics model reported AUC 0.80 on one held-out patient set (Christodoulou et al., 2026), and a nested-CV model of volume increase at the next scan reported 0.728 (Rai, 2026) [CHECK: Rai 2026 is "repo, unverified (this pass)" in COMPARABLE_RESULTS.md]. An earlier internal report on the same dataset, which we call the PRISM report, also described a radiomics model (the PRISM report model) with test AUC 0.80 on one 30-patient split (the PRISM report split). That model is the starting point of this study.

Three problems are well documented in this literature. First, single train–test splits are unstable in small radiomics cohorts. Over 1,000 random splits of a brain-tumour radiomics dataset, one split gave 0.882 in training and 0.667 in testing, and another 0.709 and 0.911 (An et al., 2021). Cross-validation error bars are about ±10% at n = 100 (Varoquaux, 2018). Second, performance drops on external data: 81% of externally validated deep-learning algorithms lost performance, and 24% lost at least 0.10 (Yu et al., 2022), and only 29.4% of neuro-oncology radiomics studies were externally validated (Park et al., 2020). Third, clinical-only comparators are rarely reported. Where they are, they are competitive. In pseudoprogression classification, clinical variables with MGMT reached 0.69 and adding radiomics made the model significantly worse (Leone et al., 2023). A clinical-only model reached 0.727 externally against 0.717 for

imaging alone (Gomaa et al., 2025). In survival prediction from pre-treatment MRI, clinical variables reached 0.74 against 0.75 for the combined radiomics model (Abler et al., 2025). These are different tasks, often with diffusion or perfusion inputs, so their AUCs are context rather than like-for-like comparators. One study found the reverse: a clinical-only random forest scored 0.52 on a 19-patient test set (Jang et al., 2018).

A fourth problem has received little attention. In the two best-documented clinical comparators, time since the end of radiotherapy was an input, and in one it was the most influential feature (Gomaa et al., 2025; Leone et al., 2023). A surveillance cohort mixes scans taken before radiotherapy, during it, in the 12-week pseudoprogression window and in later follow-up. Event rates differ sharply between these phases. A model that only recognises the phase can rank scans well when the test set contains many easy phases. We found no study that treats phase of care as a confounder of imaging-model AUC for forward progression prediction. A related issue is leakage of information that is not yet available at prediction time (Kapoor & Narayanan, 2023). In routine clinical tables, treatment fields are recorded once per patient. If they are read as of the end of follow-up, a scan taken before radiotherapy "knows" the radiotherapy that follows it, and the model learns the phase of care from the future.

We therefore re-evaluated the PRISM report model and asked four questions. (i) What does a single split of this size do to the AUC, and why was the PRISM report split favourable? (ii) Under repeated, patient-grouped, censoring-aware evaluation, do MRI radiomics or frozen deep features add to a clinical model that knows where each scan sits in the treatment course? (iii) Does the answer change within the post-radiotherapy surveillance phase, with a cleaner imaging-defined label, with a consistent segmenter, on an external cohort, or with diffusion imaging? (iv) If the imaging gain is absent, what can an auditable clinical model offer at AUC about 0.70? Our contributions are a split-variance and composition analysis of the single split, a head-to-head of clinical and imaging features under nested evaluation with internal and external testing, the identification of phase of care as the driver of discrimination, and an auditable decision-support layer whose value and limits we test against pre-specified criteria.

2 Methods

2.1 Study design and reporting

This retrospective study of public, de-identified data develops and internally evaluates prediction models on one cohort (MU-Glioma-Post), evaluates locked models on a second (LUMIERE) and runs a supporting analysis on a third (CFB-GBM). Reporting follows TRIPOD+AI (Collins et al., 2024) and CLEAR (Kocak et al., 2023 [CHECK]); both checklists are cross-referenced in Supplementary S17. Analyses labelled *pre-specified* had their plan and decision rule written into the analysis document and code before any model output of that analysis was seen; for

the clean-label, audit and CFB-GBM analyses a dated plan snapshot was saved before the run (Supplementary S1). Everything else is secondary or exploratory.

2.2 Data sources

MU-Glioma-Post (development; Mahmoud et al., 2025) is a single-centre collection of post-operative MRI (T1, T1c, T2, FLAIR) with four-compartment masks (non-enhancing tumour core, NETC; surrounding FLAIR hyperintensity, SNFH; enhancing tumour, ET; resection cavity, RC) and a clinical workbook. The workbook gives days from diagnosis to each MRI, surgery, RT and chemotherapy start and end, first progression and death, plus age, sex, grade, IDH, MGMT and the type of first progression (imaging, clinical or both). Images carry no acquisition metadata. **LUMIERE** (external; Suter et al., 2022) is a single-centre longitudinal glioblastoma collection with an expert RANO rating per follow-up, automatic DeepBraTumIA segmentations, IDH and MGMT, but no RT or chemotherapy dates; its only treatment-time variable is a "< 3 months after RT" flag. **CFB-GBM** (supporting; TCIA doi:10.7937/v9pn-2f72 [CHECK]) contains 264 patients treated with the Stupp protocol, imaged at RT planning (t0) and twice afterwards (t1, t2), including ADC, with volumetric RANO 2.0 labels and precomputed PyRadiomics features on the gross tumour volume (GTV), but no IDH or MGMT. We used only its released tables.

2.3 Index scans and outcome

An index scan is a post-treatment MRI before first documented progression and before late treatment, with all four sequences and a mask, a known and plausible MRI day, and no recorded death before it. The flow from 596 timepoints of 203 patients to 308 index scans of 155 patients is in Supplementary S2. The grade 4 subset (249 scans, 126 patients) is the pre-specified primary population. LUMIERE index scans are rated follow-ups at least two weeks after surgery, before the first rating of progressive disease. Each patient contributes all eligible scans, and every analysis keeps a patient's scans together.

The primary outcome is documented first progression within $H = 120$ days of the index scan (90 and 180 days as sensitivity analyses). A scan is negative only if progression-free follow-up reaches 120 days; otherwise it is censored. Death without documented progression censors at the death day. Rules for conflicting dates were fixed in an audit before the final run; six index scans left the cohort, and at 120 days 17 negatives became censored and one became positive (Supplementary S3). In LUMIERE, progression is the first expert rating of progressive disease. The PRISM report used a "naive" label that counted censored scans as negatives; we use it only where we reproduce the PRISM report. Models are fitted on scans whose 120-day status is known. Discrimination is the inverse-probability-of- censoring-weighted (IPCW) cumulative/dynamic AUC at 120 days (Uno et al., 2007; Blanche et al., 2013). In a planned comparison of label

formulations, none changed discrimination (all 36 ΔAUC CIs included 0), but the naive label under-predicted risk (observed/expected, O/E, 1.12–1.19 against 0.95–0.99; Supplementary S3).

2.4 Predictors

All predictors are defined as of the scan day (definitions in Supplementary S4).

- **Clinical**: age, sex, IDH (wild type, mutant, unknown), MGMT (methylated, unmethylated, unknown).
- **Timing**: days from diagnosis, days since RT end, an early post-RT flag (0–84 days after RT end, the RANO 12-week window) and its interaction with MGMT methylation, visit index, and an unknown-RT-end flag.
- **Treatment, as of the scan**: whether RT and chemotherapy had started, whether RT was completed, prescription, RT start day, RT duration, surgery day, biopsy before resection and previous brain tumour. A treatment fact enters only once its documented start day is on or before the scan day; RT duration and days since RT end appear only after RT has ended. Unknown values are imputed in-fold with missingness indicators. The **baseline** model is clinical + timing + treatment (23 predictors, 31 design columns).
- **Phase of care**: pre-RT, during RT, early post-RT ($\leq$ 84 days) or later post-RT, plus log days since RT end.
- **Imaging measurements**: compartment volumes; morphology (ET fraction, multifocality, ET-to-cavity distance, intensities relative to normal brain); and past-only change from the previous scan and the nadir.
- **Radiomics**: the PRISM report's legacy T1c + FLAIR features (2,446 columns) and corrected features (whole-brain z-score, fixed 32-bin discretisation, per compartment, all four sequences). In every training fold each radiomics and deep feature is clipped to that fold's 1st and 99th percentiles, which removes a PyRadiomics sentinel value (10^6) found in three legacy columns of four scans (Supplementary S5).
- **Deep features**: frozen SwinUNETR encoders with BraTS 2021 and BrainSegFounder weights (Cox et al., 2024 [CHECK]), and the same architecture with random weights as a control; mask-weighted pooled features per region (Supplementary S6).

2.5 The PRISM report model, its split and the as-of-scan correction

The PRISM report model is an L2-penalised logistic regression on 46 legacy radiomics and clinical-code columns with Platt calibration; its features came from an L1 + permutation consensus ranking and its hyperparameters from an Optuna search (Supplementary S7). The PRISM

report split came from the report's own split search: for a seed, 20,000 random draws of 30 patients, keeping the draw closest to 96 test scans and 53 positives; seed 62 gives the split. We reproduced the shipped model's stored test predictions to 1.1×10^{-15}. In the label-fixed cohort the split has 126 training and 29 test patients with 82 test scans (22 naive positives). On it we fitted, once and without tuning on the split, a baseline ridge, a baseline gradient-boosting model, the baseline plus legacy radiomics, and a clinical-only ridge. CIs and paired ΔAUC against the baseline ridge come from a 1,000-draw patient-cluster bootstrap over test patients.

In earlier runs, treatment and timing features read the workbook as of the end of follow-up. A pre-RT scan therefore carried the dose, start date and "RT given" status of a course that had not started, and a "days since RT end" computed from a future date (RT status was wrong in 23 all-glioma scans and RT duration or days since RT end in 29; Supplementary S4). We fixed this with the as-of rule above and report the split before and after the fix (Table 2).

2.6 Split variance

We ran the PRISM report's split search for seeds 0–499, giving 500 distinct 30-patient test sets (82–94 scans, 10–33 positives). On each split's training patients we refitted (a) the PRISM report pipeline with its feature ranking redone (the honest estimate; Optuna not re-run), (b) the shipped 48 features (optimistic, because they were selected on patients who are test patients in most other splits), (c) the baseline ridge and (d) the baseline plus legacy radiomics. We report naive-label AUC (500 splits) and IPCW AUC (475 splits; undefined on 25), the percentile rank of the PRISM report split, and the relation of per-split AUC to test-set composition. This analysis is descriptive; verifier checks are in Supplementary S8.

2.7 Repeated cross-validation and model comparison

All repeated-CV analyses share one set of 5×5 repeated patient-grouped folds stratified by 120-day status and grade. Every fitted step — imputation, clipping, scaling, selection, stacking and hyperparameters — is fitted inside the training fold (Varma & Simon, 2006; Cawley & Talbot, 2010). The headline estimate is the mean over the five repeats of the per-repeat pooled out-of-fold IPCW AUC(120 d); an earlier repeat-averaged estimator, which acts as an ensemble, was replaced before the final run (Supplementary S9). 95% CIs and paired ΔAUC come from a 1,000-draw patient-cluster bootstrap in which all models are scored on the same resampled patients.

The primary learner is ridge logistic regression with the penalty chosen by inner patient-grouped log-loss; regularised gradient boosting is secondary. Radiomics enters through 70 configurations (feature variant $\times$ stability filter $\times$ selection $\times$ concatenation or a stacked radiomics score; Supplementary S5). The honest estimate chooses one configuration per outer training fold

by inner 3-fold patient-grouped CV (nested selection). Deep features use the same protocol over 32 configurations, with a 16-configuration random-weight control. With 70–80 uncorrected intervals, about 1.75–2 single configurations are expected to exclude 0 by chance. **The pre-specified primary development analysis** was grade 4: baseline ridge versus baseline + radiomics by nested selection.

2.8 Phase of care (pre-specified)

A ladder of non-imaging bases — R0 clinical; R1 + timing; R2 clinical + phase; R3 baseline — received four imaging additions: I1 volumes; I2 all non-radiomics imaging measurements; I3 radiomics (nested); I4 deep features (nested). Cohorts were the full cohort, the post-RT cohort (RT documented and completed; grade 4: 167 scans, 84 patients, 83 events) and the later post-RT cohort (> 84 days after RT end). Models were trained and evaluated within each cohort. The primary contrast was grade 4, post-RT, ridge, R2 + I2 versus R2, with a 100-permutation test that moves the I2 block between patients with the same number of scans; the key secondary contrast was R2 + I3 versus R2. The plan was written into the code before the full run but not committed separately (Supplementary S11).

2.9 Label, positive controls and re-segmentation

Imaging-defined label (pre-specified). Clinical-only first progressions were censored at the progression date. Rule: the label is "the bottleneck" if, in grade 4 (ridge), imaging alone or baseline + imaging (volumes + change) beats the baseline by $\Delta\text{AUC} \geq 0.03$ with a 95% CI excluding 0. **Positive controls.** We checked volumes against the dataset's own volume table, label mapping, label–date alignment and 12 random masks, and measured concurrent detection of the progression scan against verified 120-day negatives (imaging only). **Re-segmentation (pre-specified).** All 594 complete MU scans were re-segmented with the MONAI BraTS 2021 SwinUNETR (primary) and BrainSegFounder. The new masks "help" if concurrent detection rises by ≥ 0.05 with a CI excluding 0, or if ET is larger at the progression scan than at the previous scan in more than 65% of pairs.

2.10 External evaluation and diffusion

Models trained on all MU scans were applied unchanged to LUMIERE. Without treatment dates, the transportable features are clinical + timing (log days since surgery, the $<$ 3-month flag) $\pm$ harmonised volumes; the pre-specified external model was clinical + timing + volumes trained on grade 4. We report pooled IPCW AUC, O/E and calibration slope, and on 5×5 LUMIERE folds recalibration, fine-tuning and a cohort-normalised volume sensitivity analysis (Supplementary S13). The shipped PRISM report model was applied to legacy features re-

extracted on LUMIERE.

In CFB-GBM (pre-specified; one row per patient), the outcome was volumetric RANO 2.0 progression from t1 to t2. Nested feature sets were (a) clinical, (b) + GTV volume and its change from t0, (c) + conventional T1-Gd and FLAIR radiomics, (d) + ADC radiomics, fitted by ridge with in-fold clipping and selection and evaluated by 20 repeats of 5-fold CV. Rule: ADC helps if (d) − (c) ≥ 0.03 with a CI excluding 0, with a 1,000-permutation test of the ADC block. Caveats were fixed in advance: no molecular markers; t1 falls inside the 12-week window; and volume at t1 is part of the label, so (b) is partly circular.

2.11 The auditable decision-support layer

The system wraps the baseline ridge model; it adds no new predictor (plan snapshot saved before the run; Supplementary S15). It has five parts. (1) A **locked model** with a model card: code version, hashes of every input table, each input's definition, unit, source column and as-of rule, training counts, cross-validated performance, intended use and out-of-scope uses. (2) A **per-scan audit record**: each input with its source and as-of day and a check that it was known by the scan day; phase of care; risk with a 95% interval from 200 patient-cluster bootstrap refits; exact additive log-odds contributions by group, checked against the logit; a split-conformal prediction set; and referral flags with reasons — (i) early post-RT window, (ii) out of distribution (IDH or MGMT unknown, or an input outside the training 1st–99th percentile), (iii) uncertain (conformal set $\{0, 1\}$ at $\alpha = 0.2$). (3) **Selective prediction**: risk–coverage curves compared with random removal and with composition-matched removal (same retained count per phase). (4) A **shortcut and bias audit**: refits without the "unknown" molecular indicators, timing-only and treatment-only models, subgroup calibration and a transport audit on LUMIERE. (5) **Explanation stability** and a reproducibility manifest. Everything was evaluated inside the shared folds with every fitted step, including the conformal calibration split (70/30 of training patients), fitted on training patients only.

Two quantities were pre-specified. **P1:** conformal coverage at $\alpha = 0.2$ on verified test scans ≥ 0.78 overall, with per-phase coverage reported as conditional coverage. **P2:** AUC on scans retained at about 70% coverage minus the AUC of composition-matched retention, met if its 95% CI lies above 0.

2.12 Further analyses and software

The Supplement reports a feature-group search with a permutation control (S10), literature baselines (including a Hashim-style Cox model; Hashim et al., 2026), hyperparameter grids, class imbalance, schedule coupling, feature conflicts, SHAP, calibration and decision curves (Van Calster et al., 2019; Pencina et al., 2008) and sample size by the Riley criteria (Riley et al., 2020).

Code is Python (scikit-learn, scikit-survival, PyRadiomics, MONAI, SHAP); every stage is one command with fixed seeds, and logs, folds and hashes are kept with the outputs (Supplementary S16).

3 Results

3.1 Cohorts

MU-Glioma-Post contributed 308 index scans from 155 patients (Table 1). Most were grade 4 (126 patients, 81%), IDH wild type (77%) and treated with RT (66%). At 120 days, 100 scans were followed by documented progression, 156 were not and 52 were censored (grade 4: 93 / 117 / 39). Of the grade 4 scans, 76 were taken before RT, 6 during RT, 98 in the early post-RT window and 69 later. Event rates differed by phase: in the full cohort, 8.7% of pre-RT scans were followed by progression within 120 days. LUMIERE contributed 132 index scans from 61 glioblastoma patients (47 / 76 / 9 at 120 days). IDH status was unknown in 36% of LUMIERE patients and 8% of MU patients.

3.2 The PRISM report split

On the PRISM report split, every model scored high (Figure 1A; Table 2). The PRISM report model reached 0.802 on the naive label and an IPCW AUC of 0.817 (95% CI 0.690–0.920). A ridge model with no imaging — clinical, timing and as-of-scan treatment — reached 0.871 and 0.851 (0.735–0.947). The paired difference between the two was −0.034 (−0.186 to +0.109). Adding legacy radiomics to the ridge model changed it by −0.006 (−0.081 to +0.090). A clinical-only ridge model, with age, sex, IDH and MGMT alone, scored 0.791 and 0.782 (0.638–0.897). All intervals were wide and overlapping.

How a 0.70 model scored 0.90. Before the as-of-scan correction, the same clinical + timing + treatment design fitted by gradient boosting scored 0.901 on the naive label on this split (30 test patients, 84 scans; Table 2, history rows). Under repeated CV on the same pre-correction inputs, that model averaged 0.723 (0.633–0.806) in grade 4 and 0.743 (0.666–0.816) in all gliomas. Two things lifted it. The first was leakage: pre-RT and on-RT scans carried the RT start date, dose and end date of a course that had not yet started or ended. After the correction, the boosting model scored 0.869 on the same split, and with the final labels and the 29-patient test set, 0.844 (IPCW 0.824, 0.690–0.916). Its final repeated-CV AUC is 0.687 (0.601–0.768) in grade 4 and 0.718 (0.642–0.792) in all gliomas. The ridge model moved less: 0.870 before the correction, 0.880 after it and 0.871 with the final labels. The second lift was the split itself.

Why the split is easy. Of the 60 negative test scans, 36 were taken before RT started, and none of the 36 pre-RT test scans was followed by progression (Figure 1B). In the whole cohort

9% of such scans were. So 60% of the positive–negative pairs compared a progression scan with an early post-operative, pre-RT scan. The baseline ridge ranked these pairs with AUC 0.962 and the remaining pairs with 0.735; the PRISM report model scored 0.823 and 0.771. Among the 500 design splits, the PRISM report split was at the 89th percentile for the share of pre-RT scans, the 5th for IDH-mutant scans and the 6th for later post-RT scans. A no-training rule, "the scan is post-RT", scored 0.740 on this test set (92nd percentile). On the 42 post-RT test scans (19 patients), the PRISM report model scored 0.770 (0.604–0.907) and the baseline 0.718 (0.481–0.880) (Figure 1C). The clinical-only model was nominally highest there, at 0.818 (0.664–0.924), with a difference from the baseline of +0.100 (−0.015 to +0.293): another illustration of single-split noise.

3.3 What a single split does: 500 splits of the same design

When the PRISM report pipeline re-selected its features on each split, its AUC averaged 0.720 ± 0.055 (2.5–97.5% range 0.610–0.819) (Figure 2A). Its re-run on the PRISM report split scored 0.801, above 92% of splits. Only 7% of splits reached the reported 0.802. Refitting the shipped 48 features gave 0.815 ± 0.044, but this is an upper bound: those features were chosen on patients who are test patients in most other splits. Per split, the honest version was 0.095 ± 0.040 lower than the optimistic one. Re-running the full hyperparameter search did not rescue the pipeline (Supplementary S8).

The baseline ridge averaged 0.801 ± 0.052 over the same splits, and the PRISM report split ranked it at the 91st percentile. Adding legacy radiomics gave 0.799 ± 0.053, a mean difference of −0.002; radiomics helped in 51.6% of splits. Per-split AUC tracked the phase mix of the test set (Figure 2B): for the baseline, the Spearman correlation with the AUC of the "post-RT" rule was +0.57. From composition alone, nine variables predicted 0.851 for the PRISM report split for the baseline (observed 0.871, mean 0.801; CV R^2 0.42) and 0.773 for the PRISM report pipeline (observed 0.801, mean 0.720; CV R^2 0.10). About two thirds of the split's excess over an average split is thus explained by its composition, and the rest by the particular patients.

These AUCs use the naive label and the PRISM report's case table, so their level is not comparable with repeated CV; for the same models, single-split IPCW AUCs were 0.10–0.16 above the all-glioma repeated-CV values (Table 2).

3.4 Clinical versus imaging features under repeated cross-validation

In grade 4, clinical features alone reached 0.585 (0.496–0.666). Adding timing raised this to 0.644 (0.560–0.721), and adding as-of-scan treatment to 0.698 (0.615–0.774), the baseline (Figure 3A). In all gliomas the steps were 0.619, 0.693 and 0.720 (0.638–0.792). Boosting did not beat ridge (grade 4 0.687; all gliomas 0.718). Tuning changed the baseline by at most ±0.025, and

imaging-only literature baselines were weaker (for example, a volumetric RANO-style rule 0.551; Supplementary S12).

Radiomics (pre-specified primary analysis). With nested selection over 70 configurations, radiomics changed the grade 4 baseline by −0.032 (−0.084 to +0.017; AUC 0.666) and the all-glioma baseline by +0.028 (−0.017 to +0.071; AUC 0.747) (Table 3; Figure 3B). No grade 4 configuration had a CI above 0 (8 of 70 were below 0). In all gliomas 1 of 70 did — legacy PCA-10, stacked, +0.035 (+0.002 to +0.068) — fewer than the 1.75 expected by chance. Nested radiomics models were over-confident (calibration slope 0.60 in grade 4 against 0.89 for the baseline).

Deep features. Frozen pretrained encoders, nested over 32 configurations, changed the baseline by −0.004 (−0.069 to +0.063) in grade 4 and +0.011 (−0.032 to +0.056) in all gliomas. The pretrained features did carry image-only signal (grade 4 image-only AUC 0.652, against 0.566 with random weights), but it correlated with tumour volume (r = 0.58) and was not additive. The random-weight control was worse than the baseline: −0.049 (−0.085 to −0.014).

Credit is not value. In models that included volumes, morphology and change features, these imaging groups took 45% (ridge) and 51% (boosting) of the mean absolute SHAP credit, yet the models did not improve on the baseline (0.692 and 0.694 against 0.698) (Figure 3C). In the baseline itself, treatment took 58% of the credit, clinical 25% and timing 17%; the leading features were whether RT had started and the RT start day.

3.5 Phase of care

A compact four-level phase block recovered most of the baseline's gain over clinical features: grade 4 R0 0.585, R2 0.664, R3 0.698 (R2 − R0 +0.080, +0.005 to +0.152) (Figure 4A). With phase known, volumes did as well as the full treatment block (R2 + I1 0.696) but no better. Within the post-RT cohort no model discriminated well. Grade 4 bases scored 0.549–0.604, and phase itself carried nothing (R2 − R0 −0.003).

Primary contrast (pre-specified). In grade 4 post-RT scans, adding all non-radiomics imaging measurements to clinical + phase changed AUC from 0.549 (0.458–0.642) to 0.610 (0.529–0.688): ΔAUC +0.061 (−0.038 to +0.156), permutation p = 0.21 (Figure 4B). The permutation null itself averaged +0.028: adding 44 permuted columns to a weak base raises its AUC. The key secondary contrast, radiomics, gave +0.061 (−0.041 to +0.157) with calibration slope 0.44. **The hypothesis that imaging adds value within the post-RT phase was not supported.** On top of the full baseline, every imaging block changed post-RT AUC by +0.005 to +0.04 with all CIs including 0. In an exploratory check, age alone scored 0.586 in grade 4 post-RT scans, above R2, so part of the +0.061 reflects a weak base. In the later post-RT cohort (69 scans) the bases fell below 0.5 (0.343–0.472), so contrasts there are not interpretable.

3.6 Label, positive controls and segmentation

Imaging-defined label (pre-specified). Clinical-only progression accounted for 12 of 93 grade 4 events (10 patients). With these censored, baseline + imaging reached 0.722 (0.639–0.799) against 0.696 (0.610–0.776) for the baseline: ΔAUC +0.027 (−0.033 to +0.082). Imaging alone was 0.068 below the baseline. **The rule was not met**; the documented-progression label was not shown to be the bottleneck.

Positive controls. No pipeline error was found. Our volumes reproduced the dataset's own table (Pearson r 0.996– 0.9998 by compartment), labels were aligned with scan dates and the 12 inspected masks were correct. Yet imaging detected the progression scan itself against verified negatives with AUC only 0.68 (0.62–0.74). ET volume was larger at the progression scan than at the previous scan in 49% of 74 pairs ($p = 0.54$); whole-lesion growth from the nadir was larger in 79% ($p = 0.0003$). Volume series were noisy: between scans ≤ 14 days apart the median ET change was 3.2-fold. The 120-day label largely meant "the next scan is the progression scan" (agreement 79%, κ 0.53; Supplementary S14).

Re-segmentation (pre-specified). The BraTS 2021 SwinUNETR agreed with the MU masks (median Dice ET 0.80, whole tumour 0.85) and removed part of the short-interval noise, but primary detection fell from 0.681 to 0.665 (Δ −0.016, −0.057 to +0.025), and ET was larger at progression in 45% of pairs. BrainSegFounder gave 0.677 and 41%. **The rule was not met.** In the forward 120-day task the same clinical + phase + imaging model changed by +0.030 (−0.008 to +0.069) in grade 4 with the new masks (Table 3).

3.7 External and third-cohort evaluation

On LUMIERE the pre-specified locked model (clinical + timing + volumes, trained on MU grade 4) scored 0.608 (0.469–0.739) with ridge and 0.620 (0.495–0.733) with boosting, against 0.667 and 0.675 in MU CV. Clinical + timing alone scored 0.610 and 0.632; adding volumes did not help (ridge −0.001, −0.073 to +0.057). The ridge model under-predicted risk (O/E 1.33); boosting was calibrated in the large (O/E 0.99). Recalibration brought O/E to 0.93–0.97 without changing the ranking, and no fine-tuning arm beat the locked model. Rank-normalising volumes within each cohort raised the boosting model to 0.659 (0.529–0.767), suggesting that part of the loss is segmenter scale. Frozen deep features added +0.050 to +0.089 (one of four CIs excluded 0), but a random-weight encoder gained up to +0.045, and deep features identified the cohort with AUC about 0.8. The shipped PRISM report model scored 0.613 (0.500–0.729) on 118 LUMIERE scans. In LUMIERE the time to the next scan alone scored 0.788 (0.703–0.868), higher than any locked model (Supplementary S13).

CFB-GBM (pre-specified). In 77 patients with ADC (19 events), adding ADC features to conventional radiomics changed AUC by +0.007 (−0.019 to +0.035; permutation $p = 0.151$;

minimal detectable difference 0.038). **The rule was not met.** Clinical features alone were at chance (0.501 in the 107-patient conventional cohort, 0.565 in the ADC cohort). GTV volume and its change added +0.185 (+0.064 to +0.293) and +0.148 (+0.031 to +0.266), but this block is partly circular with a label defined by the same volume. Conventional texture added nothing beyond volume (−0.009 and −0.022).

3.8 Calibration and clinical utility

The grade 4 baseline was well calibrated in the large (weighted O/E 0.96; calibration-in-the-large −0.09, −0.42 to 0.28) and mildly over-fitted (slope 0.89, 0.50–1.43). At a 30% threshold its net benefit was 0.232 against 0.189 for treating all. Radiomics lowered net benefit at 30% by −0.020 (−0.038 to −0.004); deep features changed it by at most ±0.002 at 10–30%. The study is under-powered by the Riley criteria even for the baseline: 3.0 events per parameter for 31 design columns, against 2,190 scans (970 events) required. The radiomics result held in nine sensitivity analyses (ΔAUC −0.032 to +0.028, no CI excluding 0), and the baseline held within strata of the interval to the next scan (0.687, 0.595–0.768) and against re-balancing or XGBoost (Supplementary S12).

3.9 The auditable layer

The system's cross-validated predictions reproduced the baseline's out-of-fold predictions (maximum difference 1.7×10^{-16}), and the additive explanation reproduced every logit (maximum error 1.3×10^{-15}). An example record is in Supplementary S15.

P1 (coverage) was met overall, with conditional failures. At $\alpha = 0.2$, conformal coverage on verified test scans was 0.841 (0.790–0.885) in grade 4 and 0.823 (0.776–0.865) in all gliomas. Coverage failed in two places. Almost every pre-RT scan received the set $\{0\}$, so all 5 grade 4 pre-RT progressions (10 in all gliomas) fell outside their sets. During RT, coverage was 0.567 (6 scans).

P2 (abstention beyond case mix) was not met. At about 70% coverage, AUC on the retained scans was 0.731 (0.635–0.812) against 0.698 for random removal, but composition-matched removal gave 0.725: P2 +0.006 (−0.023 to +0.036) in grade 4 and +0.003 (−0.031 to +0.036) in all gliomas (Figure 5). Abstention kept 97.6% of pre-RT scans. The early post-RT referral rule raised AUC to 0.723 only by removing a whole phase; within-phase AUCs were 0.555 (pre-RT), 0.663 (early post-RT) and 0.597 (later post-RT), all below the pooled 0.698. Any referral flag applied to 72.6% of grade 4 scans.

The audit found four problems that the headline AUC hides. (1) A missingness shortcut: IDH was unknown in 0 of 93 progression scans but in 6 of 117 non-progression and 5 of 39 censored scans; removing the unknown-marker indicators lowered grade 4 AUC by −0.023

(−0.047 to −0.001). (2) The model is a treatment-course model: treatment features alone scored 0.701 (0.622–0.776), Δ +0.003 (−0.050 to +0.048) against the full baseline. (3) A molecular look-ahead: IDH and MGMT were used at 8 grade 4 scans taken 1–58 days before the diagnosis date. (4) The LUMIERE miscalibration came mostly from the same shortcut: IDH was unknown in 40.9% of LUMIERE scans against 4.4% in MU, and the mean contribution of the IDH-unknown indicator shifted by −0.68 logits; refitting without the unknown indicators moved LUMIERE O/E from 1.33 to 0.89. Under this shift, conformal coverage on LUMIERE fell to 0.715 (0.340 on events). Explanation signs were stable, but the exact top 5 features recurred in at most 4% of refits.

4 Discussion

We set out to re-evaluate a radiomics model that scored AUC 0.80 on one 30-patient split of a public post-treatment glioma cohort. Four findings stand out. First, the split was favourable: over 500 splits of the same design the PRISM report pipeline averaged 0.72, and 7% of splits reached the reported value. Second, under repeated, patient-grouped and censoring-aware evaluation, a model built from clinical variables and the as-of-scan treatment course reached about 0.70, and neither radiomics, frozen deep features nor a consistent re-segmentation added measurable discrimination. Third, the clinical model's signal is largely phase of care: whether and when RT started and ended. Within the post-RT surveillance phase no model, with or without imaging, discriminated well. Fourth, an audit layer around the clinical model met its coverage target and made each prediction traceable, but it did not improve discrimination beyond case mix, and its main value was to expose problems that the AUC hides.

What single splits do. The single split did more than add noise. It changed the ranking question. When 60% of the case–control pairs compare a progression scan with a fresh post-operative scan before RT, any model that recognises a pre-RT scan scores well. The clinical model does this through treatment status. The radiomics model does it through T1c texture. A no-training "post-RT" rule scored 0.740 on that test set. This mechanism is not specific to radiomics. It also lifted a no-imaging model to 0.871, and, while future treatment dates still leaked into the features, a boosting model to 0.901. That model averaged about 0.70 under repeated CV. This agrees with the instability across random splits shown by An et al. (2021) and adds a specific mechanism: when cohorts mix phases of care with very different event rates, the composition of a small test set is a major source of variance, explaining about two thirds of this split's excess. A split search that targets the number of test scans and positives does not control it. We therefore suggest four reporting rules for this task: repeated grouped CV; the distribution over splits when a single split is reported; results within phases of care; and a clinical comparator that knows the treatment course.

Phase of care as a confounder. Event rates after a surveillance MRI depend strongly on

where the scan sits in the course of treatment. A scan before RT is rarely followed by documented progression within 120 days. A scan in the early post-RT window often is, partly because of pseudoprogression and the follow-up imaging it triggers (Brandes et al., 2008; Taal et al., 2008; Wen et al., 2023). A four-level phase block recovered most of the clinical model's gain over clinical variables, treatment features alone reproduced the whole baseline (0.701), and SHAP gave treatment status the largest share of credit. Two published clinical comparators also use time since the end of RT and find it influential (Gomaa et al., 2025; Leone et al., 2023), though in a pseudoprogression task with an already suspicious lesion. Imaging models that pool pre-RT and surveillance scans can therefore report high AUCs for phase recognition rather than for progression biology. Our pre-specified test of imaging within the post-RT phase was negative ($+0.061$, -0.038 to $+0.156$), and part of its point estimate was an artefact of adding columns to a weak base.

Why imaging did not add. We found no pipeline defect: volumes matched the dataset's own table, labels were aligned and masks looked correct. Yet even at the scan where progression was documented, compartment volumes separated it from verified negatives with AUC only 0.68, and ET volume was no larger than at the previous scan in half of the pairs. Three features of the data explain this. The label is a documentation event: about one in eight grade 4 events was a clinical-only progression, and the 120-day label mostly means that the next scan was the progression scan. Volume series were noisy, with changes over days as large as over months; a consistent segmenter reduced but did not remove this. And the cohort is small: by the Riley criteria even the 31-column baseline would need about ten times more scans, and radiomics models far more (Riley et al., 2020). Cleaning the label and re-segmenting moved imaging in the expected direction by about $+0.03$, but neither met its pre-specified rule. A real imaging effect of that size cannot be excluded; our data cannot detect it. Diffusion did not change the picture in CFB-GBM, although that test was small (19 events) and used precomputed features. The task literature where imaging performs well uses diffusion and perfusion for pseudoprogression classification (Kim et al., 2019; Taylor et al., 2022), a different question from forward risk at every surveillance scan.

Credit is not value. Imaging took about half of the SHAP credit in models that included it, without any gain in discrimination. Attribution shares among correlated features describe how a model divides its output, not whether a feature group is needed; claims of imaging value need a paired difference with an interval.

External validity. Locked models lost about 0.05–0.06 AUC on LUMIERE, in line with the drops reported for externally validated imaging models (Yu et al., 2022). The shipped PRISM report model fell from 0.80 to 0.61. LUMIERE lacks treatment dates, so the treatment-aware model could not be transported, and the shared model's under-prediction came mostly from a missingness indicator: IDH status is often unknown in LUMIERE, and in MU unknown IDH occurred only in scans without progression. The time to the next scan alone predicted the

LUMIERE label better than any model, so external AUCs on this kind of label are partly schedule-dependent. In MU, by contrast, the baseline kept its ranking within strata of the scan interval.

What an auditable model offers at AUC about 0.70. We did not find a triage benefit. Abstaining on uncertain scans and referring early post-RT scans raised the AUC of the remaining scans only by changing their phase mix, and the flags together covered most scans. The conformal sets met their marginal target, but they did so mainly by assigning "no progression" to pre-RT scans, missing every pre-RT progression, and coverage broke under the shift to LUMIERE. What the layer did provide is auditability. Each record shows what the model knew at the scan date, where the scan sits in the treatment course, an interval, and an exact explanation. Running the audit found a missingness shortcut, a look-ahead in the use of molecular results before diagnosis, and the cause of the external miscalibration, none of which the cross-validated AUC revealed. We think this is the honest use of such a layer at this level of performance: to make a modest model's reasoning and failure modes visible, not to claim better decisions.

Comparison with other work. Our grade 4 baseline (0.698) is close to the clinical comparators reported for adjacent tasks (0.69 in Leone et al., 2023; 0.727 externally in Gomaa et al., 2025) and to the nested-CV result for a volume-defined label on the same dataset (0.728; Rai, 2026 [CHECK]). It is below the single-test-set 0.80 reported on MU-Glioma-Post (Christodoulou et al., 2026), whose label (progression status of each scan) differs from our forward 120-day label; we did not re-evaluate that model. Higher AUCs for forward recurrence prediction elsewhere (for example Chen et al., 2026; Tak et al., 2025) use different cohorts, horizons and units, and some fall externally (0.887 to 0.762 in Chen et al., 2026).

Limitations

The development cohort is a single centre with 126 grade 4 patients and 93 events at 120 days, far below recommended sample sizes; small imaging effects cannot be excluded. The outcome is documented progression from routine records, not a blinded per-scan RANO read, and its timing depends on scheduling. MU images lack acquisition metadata, so scanner and field-strength effects could not be audited, and race could not be studied. The external cohort is small (61 patients), uses a different segmenter and an expert RANO label, and lacks treatment dates, so the main model was not externally tested. Several analyses were added after earlier results were known (for example the clean-label and re-segmentation tests followed the imaging null), although each had its rule fixed before its own run; the phase-ladder plan was written into the code but not separately committed. Radiomics discretisation used a fixed bin count rather than the IBSI-recommended fixed bin width (Zwanenburg et al., 2020), and the ICC stability filter covered only T1c and FLAIR. Deep features were frozen; fine-tuning and diffusion-based deep features were not run. Feature pipelines in two secondary analyses (the feature-group search and one label-formulation set) still contain unclipped legacy radiomics. The shipped

PRISM report model's LUMIERE score predates the final external label definition. The audit found a small molecular look-ahead (8 grade 4 scans) that also affects the reported baseline; its size was not quantified. There was no reader study, so we cannot say whether the audit records change clinical decisions.

Conclusion

On MU-Glioma-Post, a radiomics model's AUC of 0.80 reflected one favourable split; across splits its pipeline averaged 0.72, and a clinical model that knows the treatment course reached about 0.70 under repeated cross-validation. MRI radiomics, frozen deep features, re-segmentation, a cleaner label and diffusion features added no measurable discrimination. Phase of care drives discrimination on this task and should be reported as a stratum and controlled as a confounder. An audit layer can make a modest model's knowledge and failure modes explicit, but on these data it did not make triage better than case mix.

Declarations

Ethics. [PENDING AUTHOR DECISION: secondary analysis of public de-identified datasets (MU-Glioma-Post, LUMIERE, CFB-GBM); state the institutional determination or waiver and cite the source datasets' approvals.]

Funding. [PENDING AUTHOR DECISION].

Competing interests. [PENDING AUTHOR DECISION].

Data availability. [PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83), LUMIERE (figshare; Suter et al., 2022) and CFB-GBM (TCIA, doi:10.7937/v9pn-2f72) are public; derived tables, fold assignments and locked model files — state what is released.]

Code availability. [PENDING: public repository link and release tag.]

Protocol and registration. No registered protocol. Analysis plans and dated plan snapshots are in the repository (Supplementary S1).

Patient and public involvement. None.

Use of the name "PRISM". The PRISM report is an unpublished internal report; "PRISM report model" and "PRISM report split" refer to the model and split described there.

Tables

Table 1. Cohort characteristics. Index scans are post-treatment MRIs before first documented progression. Values are n (%) or median [IQR]. Patient rows use each patient's first index scan. Full table with definitions: Supplementary Table S2.

	MU all gliomas	MU grade 4	LUMIERE
Patients / index scans	155 / 308	126 / 249	61 / 132
Index scans per patient, median (range)	2 (1–6)	2 (1–6)	2 (1–10)
Age, years	61 [47–69]	62 [53–70]	62 [53–70]
Male sex	80 (52%)	66 (52%)	32 (52%)
IDH wild type / mutant / unknown	120 (77%) / 22 (14%) / 13 (8%)	112 (89%) / 8 (6%) / 6 (5%)	38 (62%) / 1 (2%) / 22 (36%)
MGMT methylated / unmethylated / unknown	53 (34%) / 73 (47%) / 29 (19%)	45 (36%) / 62 (49%) / 19 (15%)	26 (43%) / 28 (46%) / 7 (11%)
WHO grade 4	126 (81%)	126 (100%)	61 (100%)
Radiotherapy documented	102 (66%)	89 (71%)	not recorded
Initial chemotherapy documented	96 (62%)	88 (70%)	not recorded
Days from origin to scan[a]	97 [46–186]	96 [54–182]	119 [82–226]
RT completed by scan day (scans)	185 (60%)	167 (67%)	not recorded
Early post-RT window[b] (scans)	104 (34%); unknown 94	98 (39%); unknown 65	27 (20%)
Gap to next MRI, days	77 [50–103]	74 [49–98]	91 [77–117]
Death without documented progression (scans)	22 (7%)	20 (8%)	7 (5%)
120 d: progression / no progression / censored	100 / 156 / 52	93 / 117 / 39	47 / 76 / 9
90 d	65 / 195 / 48	62 / 152 / 35	24 / 101 / 7
180 d	144 / 105 / 59	131 / 75 / 43	62 / 61 / 9

[a] MU: days since diagnosis; LUMIERE: days since first surgery.
[b] MU: 0–84 days after the documented RT end date; LUMIERE: rater's "< 3 months after RT" flag.

Table 2. The same models on a single split, across 500 splits and under repeated cross-validation. IPCW AUC(120 d) with 95% patient-cluster bootstrap CI unless stated. Single split = PRISM report split (all gliomas; 29 test patients, 82 scans). 500 splits = the PRISM report split design (naive label; for the PRISM report model, its pipeline with features re-selected per split). 5 × 5 CV = repeated patient-grouped CV on the label-fixed cohort. History rows document the as-of-scan treatment correction on the earlier 30-patient / 84-scan version of the split; "pre-fix (leaky)" means that future RT dates and RT status entered the treatment and timing features.

Model	Single split: naive AUC	Single split: IPCW AUC [95% CI]	Δ vs baseline ridge [95% CI]	500 splits: naive mean ± SD (2.5–97.5%)	5 × 5 CV, all gliomas	5 × 5 CV, grade 4
PRISM report model	0.802	0.817 [0.690, 0.920]	−0.034 [−0.186, +0.109]	0.720 ± 0.055 (0.610–0.819)	n/a (fixed model)	n/a (fixed model)
Baseline ridge	0.871	0.851 [0.735, 0.947]	–	0.801 ± 0.052 (0.694–0.895)	0.720 [0.638, 0.792]	0.698 [0.615, 0.774]
Baseline GBM	0.844	0.824 [0.690, 0.916]	−0.027 [−0.127, +0.062]	not run	0.718 [0.642, 0.792]	0.687 [0.601, 0.768]
Baseline + legacy radiomics	0.862	0.845 [0.746, 0.936]	−0.006 [−0.081, +0.090]	0.799 ± 0.053 (0.674–0.892)	0.743 [0.678, 0.803]	0.703 [0.627, 0.774]
Clinical only (ridge)	0.791	0.782 [0.638, 0.897]	−0.069 [−0.215, +0.058]	not run	0.619 [0.540, 0.696]	0.585 [0.496, 0.666]
History rows (30 patients / 84 scans; earlier cohort and inputs)						
PRISM report model	0.802	0.811 [0.694, 0.919]	not computed	–	–	–
Baseline ridge, pre-fix (leaky)	0.870	0.844 (no CI)	not computed	not run	0.719 [0.648, 0.788][c]	0.694 [0.613, 0.775][c]
Baseline ridge, after as-of-scan fix	0.880	0.860 [0.730, 0.943]	not computed	not run	0.749 [0.675, 0.820][d]	0.729 [0.648, 0.811][d]
Baseline GBM, pre-fix (leaky)	0.901	0.881 (no CI)	not computed	not run	0.743 [0.666, 0.816][c]	0.723 [0.633, 0.806][c]
Baseline GBM, after as-of-scan fix	0.869	0.855 (no CI)	not computed	not run	0.740 [0.663, 0.815][d]	0.731 [0.641, 0.813][d]

[c] Pre-fix inputs, pre-label-fix cohort, earlier repeat-averaged estimator.
[d] Intermediate inputs after the as-of-scan fix, before the label fixes; repeat-averaged estimator. History-row CV values are not comparable with the final rows.

Table 3. Does imaging add to non-imaging information? Paired ΔAUC with 95% patient-cluster bootstrap CI. Reference: the baseline (clinical + timing + treatment, ridge) unless stated in the section heading. Grade 4 (G4) is primary. * = CI excludes 0 (uncorrected; 3 of the 32 intervals in Figure 3B; the CFB-GBM row is added here).

Analysis	G4 ΔAUC [95% CI]	All gliomas ΔAUC [95% CI]
Radiomics vs baseline		
Nested over 70 configurations (pre-specified primary)	−0.032 [−0.084, +0.017]	+0.028 [−0.017, +0.071]
Legacy top-10, concatenated (single configuration)	+0.005 [−0.059, +0.067]	+0.024 [−0.026, +0.075]
Legacy PCA-10, stacked (best post hoc, optimistic)	+0.023 [−0.013, +0.057]	+0.035 [+0.002, +0.068]*
Frozen deep features vs baseline		
Nested over 32 (BraTS21 SwinUNETR + BrainSegFounder)	−0.004 [−0.069, +0.063]	+0.011 [−0.032, +0.056]
BraTS21 SwinUNETR, lesion PCA-10 stacked	+0.016 [−0.029, +0.059]	+0.017 [−0.014, +0.050]
BrainSegFounder, lesion PCA-10 stacked	+0.014 [−0.029, +0.057]	+0.024 [−0.013, +0.064]
Random-weight encoder, nested (control)	−0.049 [−0.085, −0.014]*	−0.009 [−0.032, +0.009]
Imaging-defined label (clinical-only progression censored)		
Baseline + volumes/change vs baseline (pre-specified rule)	+0.027 [−0.033, +0.082]	+0.008 [−0.036, +0.049]
Post-RT scans only		
Clinical + phase + imaging vs clinical + phase (pre-specified primary)	+0.061 [−0.038, +0.156]	+0.033 [−0.054, +0.128]
Baseline + imaging measurements vs baseline	+0.017 [−0.069, +0.093]	+0.005 [−0.075, +0.082]
Baseline + radiomics (nested) vs baseline	+0.005 [−0.070, +0.075]	+0.024 [−0.030, +0.076]
Re-segmentation (reference: same model on MU masks)		
Clinical + phase + imaging, BraTS21 SwinUNETR masks	+0.030 [−0.008, +0.069]	+0.029 [−0.010, +0.069]
Clinical + phase + imaging, BrainSegFounder masks	+0.008 [−0.023, +0.040]	+0.003 [−0.025, +0.032]
LUMIERE, locked (reference: clinical + timing)		
+ volumes	−0.001 [−0.073, +0.057]	−0.006 [−0.075, +0.055]
+ deep features	+0.057 [−0.132, +0.235]	+0.064 [−0.011, +0.144]
+ volumes + deep vs + volumes	+0.089 [−0.026, +0.215]	+0.050 [+0.002, +0.109]*
CFB-GBM (77 patients, 19 events; reference: conventional radiomics)		
+ ADC radiomics (pre-specified primary)	+0.007 [−0.019, +0.035] (permutation p = 0.151)	–

Figure legends

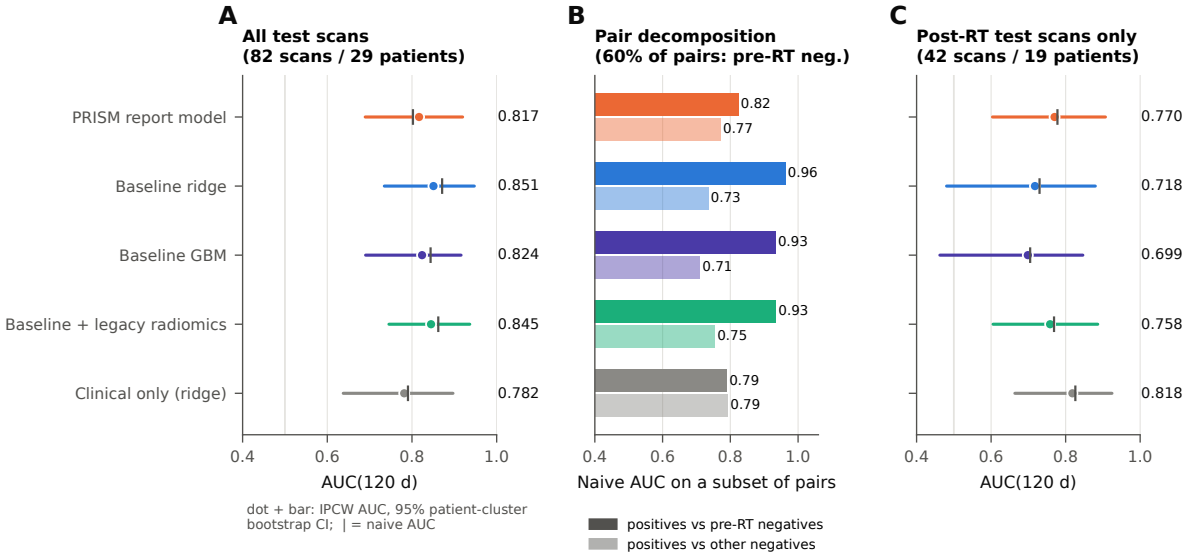


Figure 1. The PRISM report split. Discrimination on the PRISM report split (all gliomas; 126 training / 29 test patients, 82 test scans, 22 naive positives). Every model except the PRISM report model was fitted once on the training patients; the PRISM report model enters with its shipped test-set probabilities. (A) IPCW AUC with 95% patient-cluster bootstrap CI (1,000 draws; dot and bar) and naive AUC (tick). (B) Naive AUC decomposed by the type of negative scan: 36 of the 60 negative test scans were taken before RT started, and none of them progressed, so 60% of positive–negative pairs compare a progression scan with a pre-RT scan; solid bars, positives against pre-RT negatives; light bars, positives against other negatives. (C) The same models on the 42 post-RT test scans (19 patients). Intervals are wide and overlap throughout.

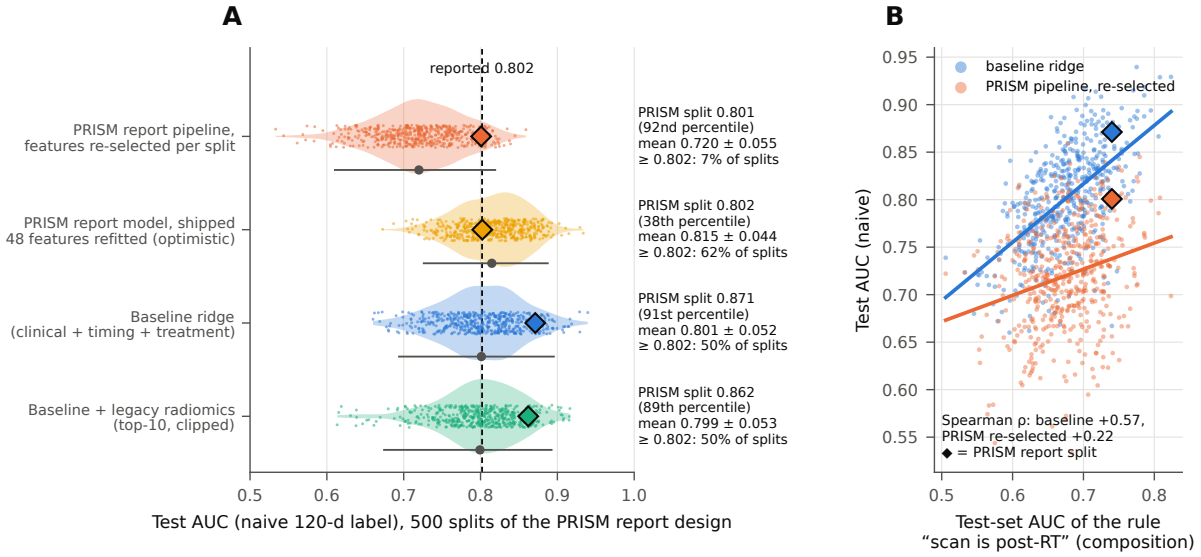


Figure 2. Where the PRISM report split sits among 500 splits of the same design. The PRISM report's split search was run for seeds 0–499; seed 62 reproduces the PRISM report split. (A) Test AUC (naive label) per split for the PRISM report pipeline with features re-selected per split (honest estimate), the shipped features refitted (optimistic upper bound), the baseline ridge and the baseline + legacy radiomics. Violin and points, 500 splits; grey dot and line, mean and 2.5–97.5% range; diamond, each model's own value on the PRISM report split with its percentile rank; dashed line, the reported 0.802. (B) Per-split AUC against the AUC of the no-training rule "scan is post-RT" on that test set (higher = more pre-RT negatives and fewer pre-RT events); lines, least-squares fits; Spearman ρ +0.57 (baseline) and +0.22 (PRISM report pipeline).

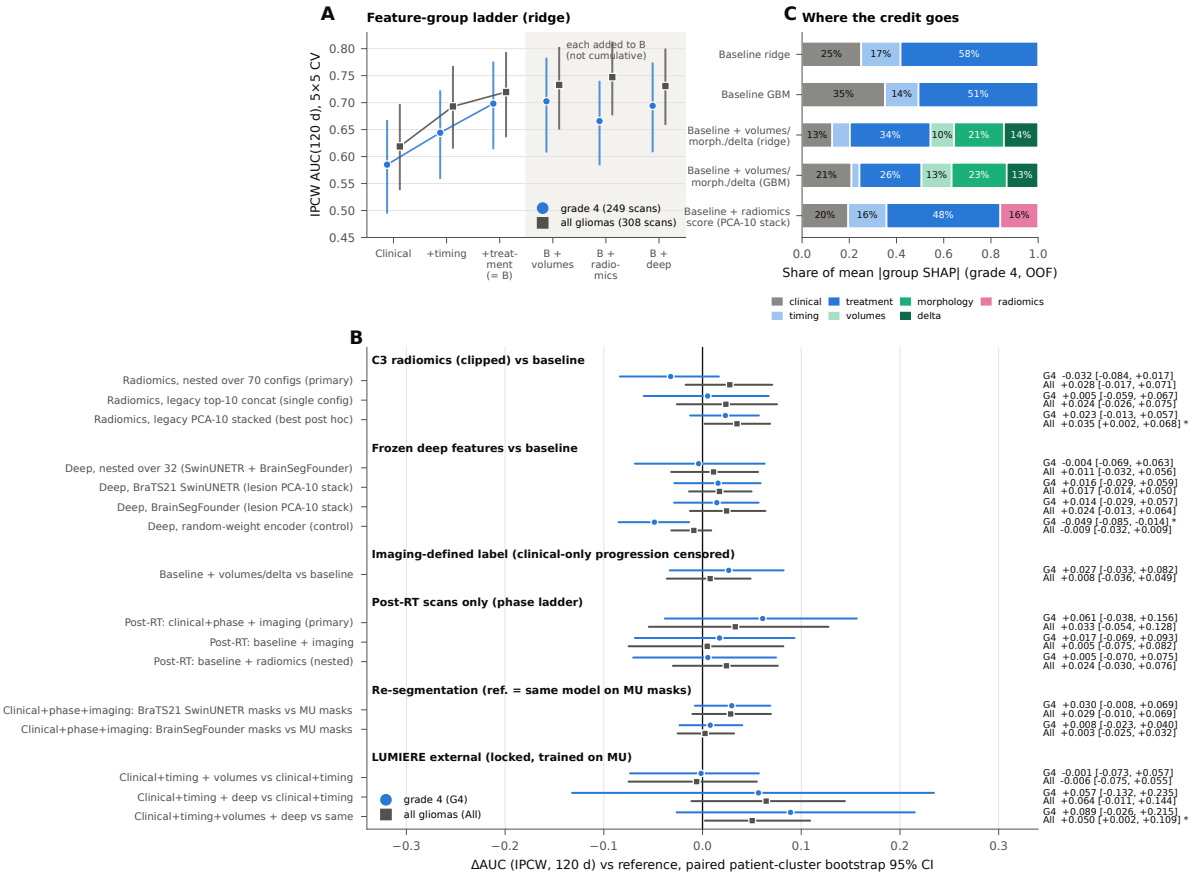


Figure 3. Clinical versus imaging features under repeated cross-validation. (A) Feature ladder (ridge; 5 × 5 repeated patient-grouped CV; IPCW AUC(120 d) with 95% patient-cluster bootstrap CI): clinical → + timing → + as-of-scan treatment (= baseline, B); the last three points each add one imaging block to B (volumes; radiomics, nested over 70 configurations; frozen deep features, nested over 32). (B) Paired ΔAUC with 95% CI against the reference named in each section (Table 3). Asterisks mark CIs excluding 0 (3 of 32, uncorrected; about 1.6 expected by chance). (C) Share of mean absolute group SHAP in grade 4 out-of-fold explanations: imaging blocks take 45–51% of the credit in the full models without improving discrimination.

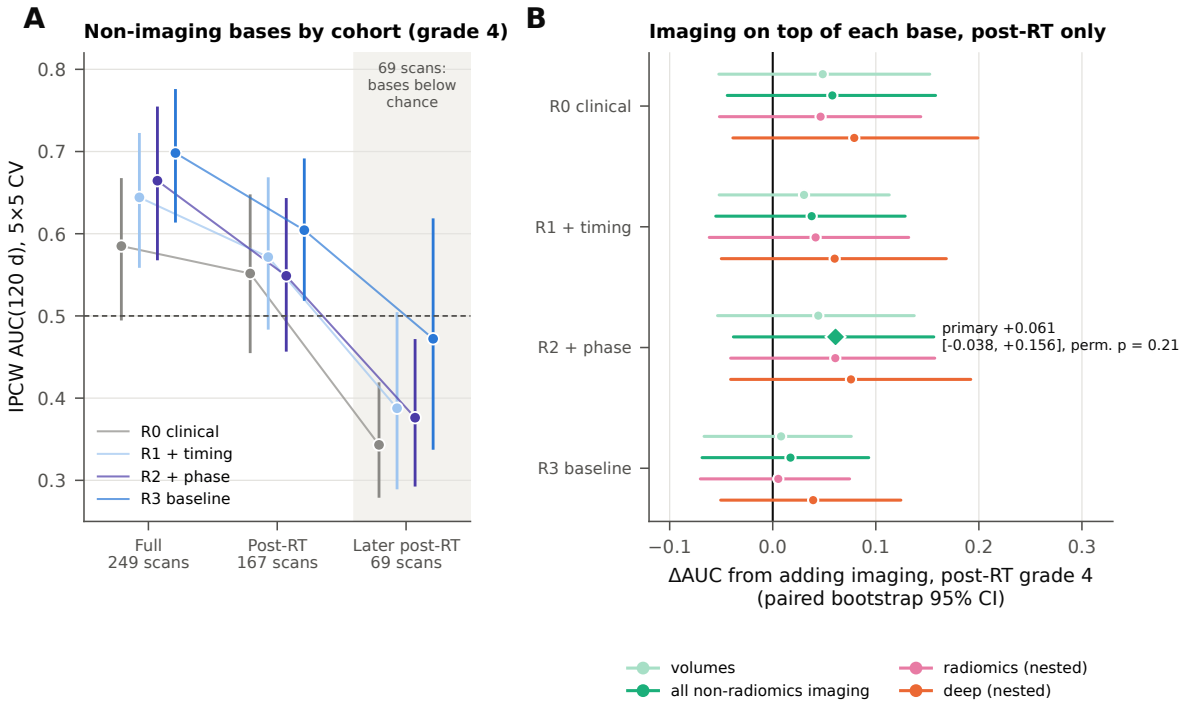


Figure 4. Phase of care. (A) Non-imaging bases (ridge, grade 4) in the full cohort (249 scans), the post-RT cohort (167 scans, 84 patients, 83 events) and the later post-RT cohort (> 12 weeks after RT end; 69 scans), each trained and evaluated within the cohort. R0 clinical; R1 + timing; R2 + four-level phase of care; R3 baseline. In the later post-RT cohort the bases fall below 0.5, the expected anti-learning of a near-null model in a small sample, so contrasts there are not interpretable. (B) ΔAUC from adding each imaging block (I1 volumes; I2 all non-radiomics measurements; I3 radiomics; I4 deep) to each base within grade 4 post-RT scans. Diamond, pre-specified primary contrast (R2 + I2 vs R2): +0.061 (−0.038 to +0.156), permutation p = 0.21; the permutation null itself averages +0.03.

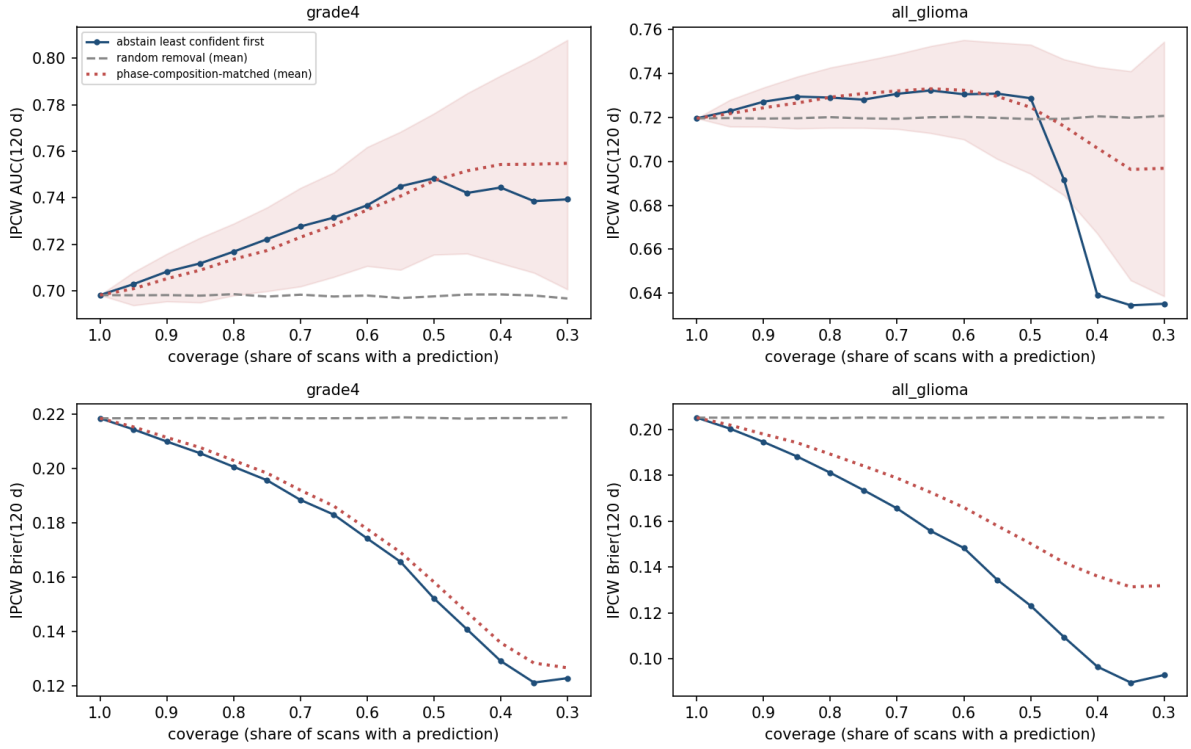


Figure 5. The auditable layer: selective prediction against case mix. Risk–coverage curves for the baseline model (5 × 5 CV, pooled out-of-fold predictions, mean over repeats), grade 4 (left) and all gliomas (right). Top, IPCW AUC(120 d); bottom, IPCW Brier score, of the scans retained when the least confident scans (lowest max(p, 1 − p)) are withheld first (solid), against random removal (dashed) and phase-composition-matched random removal (dotted; shaded band, 2.5–97.5% of the composition-matched draws). The gain over random removal is almost entirely matched by the composition-matched reference (pre-specified P2 at about 70% coverage: +0.006, −0.023 to +0.036, grade 4). [PENDING: a panel A with the system architecture and an example audit record (Supplementary S15) is to be drawn.]

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